CURRICULUM VITAE

# Muhammad Abu Bakar

Geospatial Data Specialist

Freiburg im Breisgau, Germany · imabubakar@outlook.com
linkedin.com/in/imabubakr · github.com/imabubakr · imabubakr.github.io

Visit my website

## Professional experience

11/2025 – present

**Geospatial Data Specialist** · Beagle Systems, Hamburg

Beagle Systems flies long-range drones beyond visual line of sight (BVLOS) to monitor pipelines, utilities and other infrastructure.

- Developed an open-source drone LiDAR pipeline that georeferences point clouds with PPK, classifies power-line scenes, rebuilds missing conductors from their sag and rates nearby vegetation by risk; tree-fall analysis in progress.
- Built an AI-driven right-of-way and construction-site monitoring pipeline: deep-feature-matching georeferencing, a fine-tuned foundation model for detection, an open-source VLM to verify it, and real-time inference on an NVIDIA Jetson.
- Fine-tuned a building-segmentation foundation model on in-house labels created from 10 cm orthophotos and scaled it across the Cologne area.
- Built a PostGIS and TiTiler backend that serves large GeoTIFFs to the frontend as map tiles, and supported flight planning with pre-flight checks and new geodatabase data.

03/2024 – 10/2025

**Geospatial (EO) Data Scientist, part-time** · Karuna Technology UG, Germany

- Fine-tuned a pretrained deep-learning model on in-house training data to map urban tree canopies from aerial imagery, then tracked their health with Planet NDVI and NDWI time series in Google Earth Engine to flag underperforming trees.
- Projected the long-term ecosystem services of the mapped trees through a tree-benefits API, from carbon storage to further environmental benefits, estimating the urban canopy's value over the coming decades.
- Built Google Earth Engine workflows that clean multi-year Sentinel-1 radar time series with speckle filtering and apply an equation to derive soil-moisture maps, showing where golf courses suffer from water stress.

04/2021 – 02/2024

**GIS Specialist & Data Engineer** · MapTransfer, Germany

- Managed the end-to-end geodata workflow for a construction company: georeferenced construction plans and orthophotos, converted them into cloud-optimized GeoTIFFs, uploaded them to AWS S3, and set up QGIS with PostgreSQL as the customer's secure front end.
- Designed and authored several GIS courses for students and professionals, covering LiDAR processing, spatial analysis, Spatial SQL, ALKIS and GeoServer.
- Converted AutoCAD data in QGIS into accurate, ALKIS-compliant GIS layers for telecom infrastructure projects.
- Built interactive Web GIS applications in OpenLayers, serving WMS layers through GeoServer from PostgreSQL.

## Education

2017 – 2020 **MSc Geology** · University of Freiburg, Germany

- Thesis: Simulation of multispectral and hyperspectral imaging techniques for mapping of epithermal ore deposits in vegetated zones.

2016 – 2020 **MSc Remote Sensing & Geographic Information Systems** · National University of Sciences and Technology, Pakistan

- Thesis: Development of an open-source Python-based slope stability model that incorporates remote sensing and geotechnical data for slope failure assessment.

2012 – 2015 **BSc Geology** · Bahria University, Pakistan

- Thesis: Petrophysical analysis of Joya Mair – Well 4.

## Academic projects

| | |
|---|---|
| Landsat · Sentinel-2 | Glacier mass balance of Mýrdalsjökull from historical imagery and DEMs: the melting rate doubled after 2000. |
| MODIS · TRMM | Snow cover and precipitation variability, 2000–2015. |
| Landsat 7 | Snow-cover change in the Kabul Basin and its impact on water resources. |
| Landsat 8 · ASTER | Land cover, erosion and morphometry of Mauna Kea. |

## Further training

2022 NASA Applied Remote Sensing Training (ARSET): five trainings on SAR-based disaster assessment, flood monitoring and modeling, crop mapping with polarimetric SAR and optical data, air-quality model output, and pre- and post-fire monitoring.

UNAVCO / GAGE short course: InSAR processing and time-series analysis for geophysical applications (ISCE, ARIA-tools, MintPy).

## Skills

| | |
|---|---|
| AI and machine learning | Fine-tuned foundation models, self-hosted open-source LLMs and VLMs, deep feature matching, gradient boosting; PyTorch, scikit-learn; Hugging Face Transformers, TensorFlow, SAM, XGBoost, LightGBM, CatBoost, vLLM |
| Vector data | GeoPandas, Shapely, PostGIS; ALKIS, CAD to GIS; buffers, overlays and spatial joins; GeoPackage, Shapefile, GeoJSON; Overpass API |
| Raster and remote sensing | Rasterio, GDAL, Google Earth Engine; SAR, multispectral and hyperspectral processing; xarray, cloud-optimized GeoTIFFs; ENVI, ERDAS Imagine, eCognition, ESA SNAP |
| LiDAR | PDAL, CloudCompare, WhiteboxTools; PPK, point-cloud classification; laspy, Open3D, RTKLIB |
| GIS | QGIS, ArcGIS Pro; publication-ready cartography |
| Data and cloud | PostgreSQL/PostGIS, DuckDB, AWS (S3, EC2), Docker; SpatiaLite |
| Web mapping | GeoServer, OpenLayers, TiTiler; Leaflet |
| Programming | Python, Spatial SQL, R, Bash; Matplotlib, Seaborn, Cartopy; Git |
| Geology | Numerical modeling, petrophysics |

## Languages

| | |
|---|---|
| English | C2 |
| German | B2 |

Freiburg im Breisgau, 14 October 2026

Signature